Supplement for RDD results

2026-08-31 15:07

Abstract

Quarto document renders the outputs for the full set of RDD results (sensitivity check etc). Results are also saved in /paper-outputs as a .tex fragments so they can be pulled into overleaf directly

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1 Topline

- robust effect for authors a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`)
- with authors a 2nd note (`have_another_note = note_counts[(note_counts >= 2)]`)
-
  - robust rd not sig, donut RDD sig for 2 "larger holed donuts", ordinary OLS is sig with 0.08 threshold,
- RDD is not sharp

2 Setup & Data

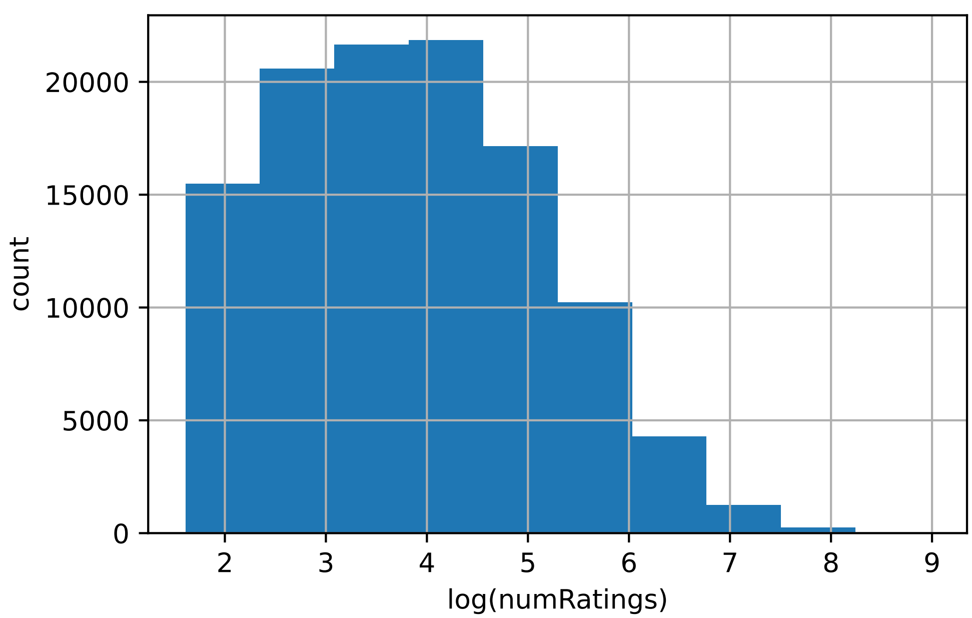
We fix the cumsum=>2 issue, so we have authors who write 2 (not 3) notes as our outcome variable We also remind the alignment filter, since this is conditioning on a collider (and makes our RDD artificially sharp)

```
Imported data for 3173276 notes
ALERT! re_auth_thresh =  2
Now we have 330234 first time note authors (all time)
Now we have 260997 first time note authors 2024-01-01 to 2026-06-02
Now we have 203392 notes written by our 3 choosen algorithms
Now we have 147890 notes classified as MISLEADING
Now we have 112758 notes which have enough ratings
--------         No filter for alignment
FINAL total 112758 Notes
of which author writes again =  77111

NB current (final), not Max, score
```

3 Diagnostic Plots

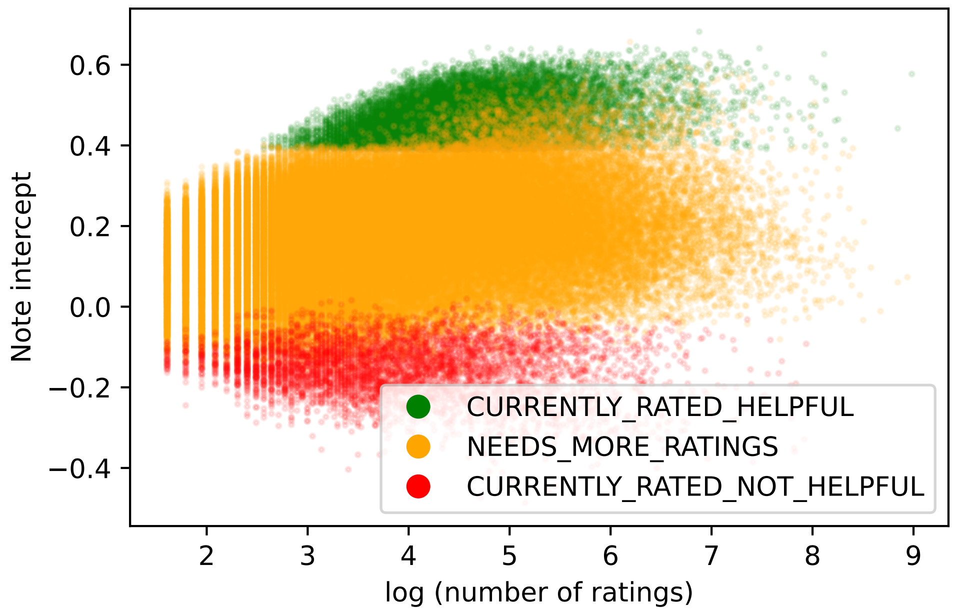
Most notes have a small number of rating, a small number have very many ratinhgs This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal



The basis for our RDD is the discontinuity between the intercept and note status (which defines publicatuon / the possiblity for publication)

Plotting log(numRatings) vs NoteIntercept, with colours according to Note final status (finalRatingStatus)

You can see the fuzzy discontinutiy - some >cutoff notes ae "NEEDS_MORE_RATINGS"



Sadly, though, our data record intercept calculated from the full set of ratings, this is not the contemporatious intercept score which determines moment-by-moment publication. It approximates it, but with contamination. Notes may be published then unpublished as the receive more ratings and their rating degrades, or there may be additional blocks on publication (e.g. requirements on total number or type of ratings, or requirements on intercept stability or a lag before publication). The CN system is not entirely clear about all these details.

These contamination can be seen by plotting indicators of ever having been published against our (final) intercept.

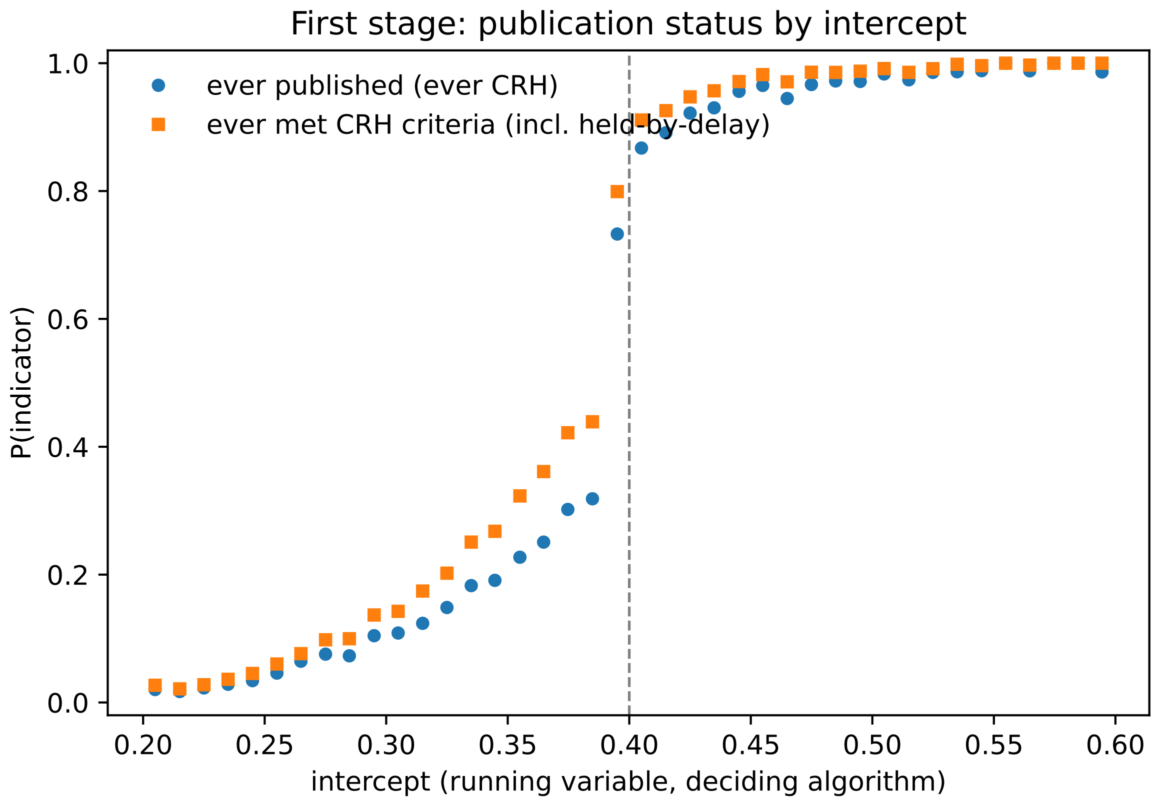
We construct an independent 'ever_published' variable from the `firstNonNMRStatus` and `mostRecentNonNMRStatus` variables. The logic being that if either of these are `CURRENTLY_RATED_HELPFUL` then the Note must have achieved CRH/published status at some point in the past (ie seperate from the finalRatingStatus score)

```
[0.35,0.40): n=  4547  P(ever_pub)=0.3851  P(ever_crit)=0.4841
[0.38,0.40): n=  1801  P(ever_pub)=0.5819  P(ever_crit)=0.6680
[0.39,0.40): n=  1145  P(ever_pub)=0.7328  P(ever_crit)=0.7991
[0.40,0.42): n=  2030  P(ever_pub)=0.8788  P(ever_crit)=0.9182

Below-cutoff P(if_written_again) by ever_published:
                    mean   size
ever_published
False           0.674867  91704
True            0.740338   4321

Below-cutoff P(if_written_again) by ever_met_crh_criteria:
                           mean   size
ever_met_crh_criteria
```

```
False                      0.676021   90438
True                       0.706819    5587
```



This shows that our secondary indicator of publication status shows a discontinuity near the cutoff, but not a sharp one

The high proportion of 'ever published' just below the cutoff suggests many notes got published and then slipped back below the threshold

We compronmise with a fuzzy RDD

4 Trad OLS RDD

Our first approach was to use OLS and an arbitary window

This does show an effect, as long as as the window is wide (0.08, it is non-significant at 0.05)

```
bandwidth =  0.08

Number of treated units: 8198
Number of control units: 8829

Estimated Treatment Effect at the Cutoff: 0.0192
I think we can intepret this as a % change the chance of authoring again if published

P-value for treatment: 0.1927
```

```
T-statistic for treatment: 1.3027
```

5 RDD using rdrobust

The rdrobust function will estimate the bandwidth from the data, as well as provide various sensitivity checks and allow a fuzzy RDD

It has the option of using a polynomial fit (which we decline), and uses all points, weigted by proximity to cutoff

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell, Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

```
running variable (x) length 112758
binary outcome (y) length 112758
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:                 110954
Polynomial Order Est. (p):                   1
Polynomial Order Bias (q):                   2
Kernel:                             Triangular
Bandwidth Selection:                     mserd
Var-Cov Estimator:                          NN

                                Left      Right
-----------------------------------------------
Number of Observations         96025      14929
Number of Unique Obs.          95875      14909
Number of Effective Obs.        4102       4723
Bandwidth Estimation           0.046      0.046
Bandwidth Bias                 0.086      0.086
rho (h/b)                      0.532      0.532
```

```
Method              Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
Conventional      -0.023    0.021   -1.111   2.664e-01    [-0.064, 0.018]
Robust                 -        -   -1.373   1.696e-01    [-0.078, 0.014]

MSE-optimal h = 0.0459

Bias-corrected point estimate: -0.0321  95% CI [-0.0779, 0.0137]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):
  h = 0.0229 (0.5x)  tau = -0.0394  95% CI [-0.1095, +0.0207]  N_L/N_R = 1993/2323
  h = 0.0459 (1.0x)  tau = -0.0233  95% CI [-0.0779, +0.0137]  N_L/N_R = 4102/4723
  h = 0.0917 (2.0x)  tau = +0.0053  95% CI [-0.0334, +0.0330]  N_L/N_R = 10844/9320
```

Mephi suggested text :

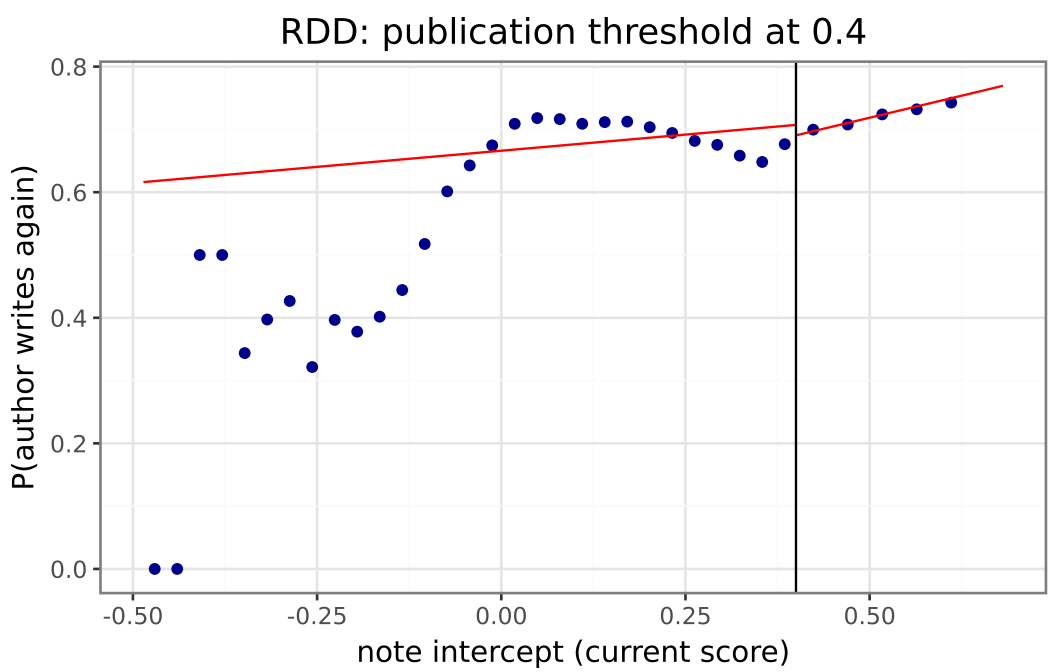
At the MSE-optimal bandwidth (h = 0.046) we estimate ˆ = -2.3pp in the probability of writing a second note (conventional local-linear estimator; robust bias-corrected 95% CI [-7.8, +1.4], p = 0.17).

6 Converting rate to baseline outcome rate

Mephi suggested text: Untreated authors at the cutoff re-author at 66.5%; treated authors at the cutoff re-author at ~64.2%. Treatment effect = -2.3pp, a -3.5% relative lift over the counterfactual baseline.

7 Showing an illustrative RDD plot

First, binned means with gloabl linear fit (this is the default plot from `rdplot`)



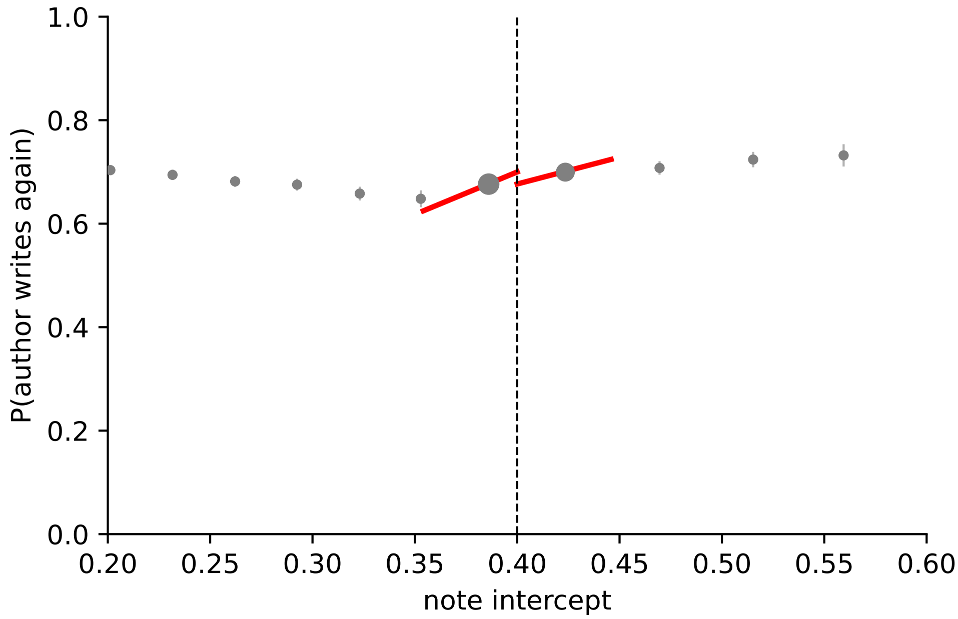
```
['rdplot_mean_bin', 'rdplot_mean_x', 'rdplot_mean_y', 'rdplot_min_bin', 'rdplot_max_bin', 'rdpl
paper-outputs/RDDrobust_scatter.png + paper-outputs/RDDrobust_scatter_data.csv written (34 bins

RD plot at c = 0.4  (p=1, triangular, N=112758)
  bins used   : 29 left / 6 right
  J_IMSE      : [29.0, 6.0]
  J_MV        : [883.0, 887.0]
  scale       : [1, 1]
  bin width   : avg [0.03054120705045503, 0.0468238353729248], median [0.030541207050455044, 0

coefficients:
       Left     Right
0  0.707181  0.690543
1  0.102853  0.280793

wrote paper-outputs/RDDrobust_scatter_meta.json, paper-outputs/RDDrobust_scatter_coef.csv
```

Now with the local-lineat fits (over the rdrobust defined window)




```
consistency check: gap at cutoff should  conventional tau
plot gap at c = -0.0233  vs  conventional tau = -0.0233
```

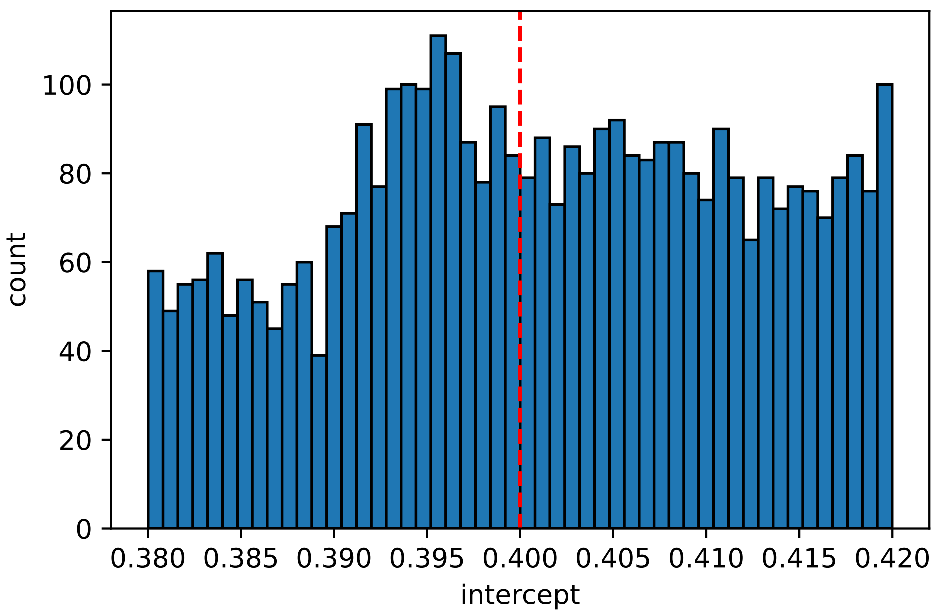
8 Robustness check - density test

Next, manipulation test/density smoothness at cutoff

if units (people) can select to be just above cutoff then density will not be smooth (think the pass grade on an exam)

```
Inspecting density with a window of +/o =  0.02
t_asy         NaN
```

```
t_jk    -4.215801
p_asy         NaN
p_jk     0.000025
dtype: float64
left     1.316348
right    0.994493
diff    -0.321855
dtype: float64
left     0.041508
right    0.040941
dtype: float64
full         110954
left          96025
right         14929
eff_left       3633
eff_right      4240
dtype: int64
leftN          [1801.0, 1957.0, 2136.0, 2343.0, 2538.0, 2731....
rightN         [2030.0, 2274.0, 2494.0, 2741.0, 2997.0, 3248....
leftWindow     [0.02, 0.022389783198856886, 0.024779566397713...
rightWindow    [0.02, 0.02232674639281102, 0.0246534927856220...
pval           [0.00022900639310549447, 1.1723340901379433e-0...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'bino_
```



A wider window shows that the density does progressively decrease up to the cutoff, but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

Exluding results at the cutoff

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
none	4102	4723	0.046	-0.0233	0.0210	0.1696	-0.0779	0.0137
±0.005	4736	5399	0.057	0.0056	0.0242	0.9339	-0.0574	0.0528
±0.010	8310	7519	0.084	0.0470	0.0221	0.0984	-0.0085	0.0996
±0.020	12875	8913	0.111	0.0493	0.0210	0.0850	-0.0067	0.1043

10 Sensitivity check - RDD sensitivity with different cutoffs

NB Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show now effect (definitely below 0.4, cutoffs above 0.4 will include some published notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that)

```
running variable (x) length 112758
binary outcome (y) length 112758
cutoff (c) is 0.3
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:                110954
Polynomial Order Est. (p):                  1
Polynomial Order Bias (q):                  2
Kernel:                            Triangular
Bandwidth Selection:                    mserd
Var-Cov Estimator:                         NN

                                Left      Right
------------------------------------------------
Number of Observations         83588      27366
Number of Unique Obs.          83460      27324
Number of Effective Obs.       13567       8704
Bandwidth Estimation           0.058      0.058
Bandwidth Bias                 0.098      0.098
rho (h/b)                      0.588      0.588

Method             Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
Conventional       0.011    0.014    0.816   4.146e-01    [-0.016, 0.038]
Robust                 -        -     0.96   3.370e-01    [-0.016, 0.046]

MSE-optimal h = 0.0576

Bias-corrected point estimate: 0.0152  95% CI [-0.0158, 0.0462]
```

```
Bandwidth sensitivity (bias-corrected point, robust 95% CI):
  h = 0.0288 (0.5x)  tau = +0.0246  95% CI [-0.0135, +0.0722]  N_L/N_R = 6371/5023
  h = 0.0576 (1.0x)  tau = +0.0111  95% CI [-0.0158, +0.0462]  N_L/N_R = 13567/8704
  h = 0.1153 (2.0x)  tau = +0.0000  95% CI [-0.0152, +0.0292]  N_L/N_R = 30038/13984
running variable (x) length 112758
binary outcome (y) length 112758
cutoff (c) is 0.35
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:                110954
Polynomial Order Est. (p):                  1
Polynomial Order Bias (q):                  2
Kernel:                            Triangular
Bandwidth Selection:                    mserd
Var-Cov Estimator:                         NN

                                Left      Right
------------------------------------------------
Number of Observations         91478      19476
Number of Unique Obs.          91333      19451
Number of Effective Obs.       21263      10492
Bandwidth Estimation           0.107      0.107
Bandwidth Bias                 0.164      0.164
rho (h/b)                      0.653      0.653

Method             Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
Conventional      -0.003    0.013   -0.208   8.353e-01    [-0.029, 0.023]
Robust                 -        -   -0.411   6.809e-01    [-0.037, 0.024]

MSE-optimal h = 0.1069

Bias-corrected point estimate: -0.0065  95% CI [-0.0372, 0.0243]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):
  h = 0.0535 (0.5x)  tau = +0.0026  95% CI [-0.0372, +0.0506]  N_L/N_R = 8631/4894
  h = 0.1069 (1.0x)  tau = -0.0027  95% CI [-0.0372, +0.0243]  N_L/N_R = 21263/10492
  h = 0.2138 (2.0x)  tau = +0.0052  95% CI [-0.0195, +0.0266]  N_L/N_R = 51948/18548
running variable (x) length 14929
binary outcome (y) length 14929
cutoff (c) is 0.45
Call: rdrobust

Sharp RD estimates using local polynomial regression.
```

```
Number of Observations:                 14929
Polynomial Order Est. (p):                  1
Polynomial Order Bias (q):                  2
Kernel:                            Triangular
Bandwidth Selection:                    mserd
Var-Cov Estimator:                         NN

                                Left      Right
------------------------------------------------
Number of Observations          5187       9742
Number of Unique Obs.           5180       9729
Number of Effective Obs.        1487       1466
Bandwidth Estimation           0.014      0.014
Bandwidth Bias                 0.023      0.023
rho (h/b)                      0.627      0.627

Method             Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
Conventional       0.062    0.037    1.705   8.826e-02    [-0.009, 0.134]
Robust                 -        -    1.659   9.711e-02    [-0.013, 0.156]

MSE-optimal h = 0.0142

Bias-corrected point estimate: 0.0716  95% CI [-0.0130, 0.1561]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):
  h = 0.0071 (0.5x)  tau = +0.0878  95% CI [-0.0244, +0.2151]  N_L/N_R = 754/778
  h = 0.0142 (1.0x)  tau = +0.0623  95% CI [-0.0130, +0.1561]  N_L/N_R = 1487/1466
  h = 0.0284 (2.0x)  tau = +0.0438  95% CI [-0.0061, +0.1132]  N_L/N_R = 2990/2851

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:                 14929
Polynomial Order Est. (p):                  1
Polynomial Order Bias (q):                  2
Kernel:                            Triangular
Bandwidth Selection:                    mserd
Var-Cov Estimator:                         NN

                                Left      Right
------------------------------------------------
Number of Observations          5187       9742
Number of Unique Obs.           5180       9729
Number of Effective Obs.        1487       1466
Bandwidth Estimation           0.014      0.014
Bandwidth Bias                 0.023      0.023
```

```
rho (h/b)                      0.627      0.627

Method             Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
Conventional       0.062    0.037    1.705   8.826e-02    [-0.009, 0.134]
Robust                 -        -    1.659   9.711e-02    [-0.013, 0.156]
```

All these are non-significant

11 Sensitivity check - data for different time windows

```
PERIOD 1, from 2025-04-08 to 2026-06-01
ALERT! re_auth_thresh =  2
Now we have 330234 first time note authors (all time)
Now we have 87791 first time note authors 2025-04-08 to 2026-06-01
Now we have 68536 notes written by our 3 choosen algorithms
Now we have 52423 notes classified as MISLEADING
Now we have 34862 notes which have enough ratings
--------         No filter for alignment
FINAL total 34862 Notes
of which author writes again =  19514

NB current (final), not Max, score
running variable (x) length 34862
binary outcome (y) length 34862
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:                34856
Polynomial Order Est. (p):                 1
Polynomial Order Bias (q):                 2
Kernel:                           Triangular
Bandwidth Selection:                   mserd
Var-Cov Estimator:                        NN

                                Left      Right
------------------------------------------------
Number of Observations         29401       5455
Number of Unique Obs.          29391       5448
Number of Effective Obs.        2460       2627
Bandwidth Estimation           0.064      0.064
Bandwidth Bias                 0.099      0.099
rho (h/b)                      0.642      0.642

Method             Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
```

```
Conventional      -0.001     0.03   -0.036   9.712e-01     [-0.06, 0.058]
Robust                 -        -   -0.253   8.000e-01    [-0.079, 0.061]

MSE-optimal h = 0.0636

Bias-corrected point estimate: -0.0090  95% CI [-0.0787, 0.0607]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):
  h = 0.0318 (0.5x)  tau = -0.0148  95% CI [-0.1174, +0.0793]  N_L/N_R = 1050/1320
  h = 0.0636 (1.0x)  tau = -0.0011  95% CI [-0.0787, +0.0607]  N_L/N_R = 2460/2627
  h = 0.1271 (2.0x)  tau = +0.0228  95% CI [-0.0381, +0.0637]  N_L/N_R = 6786/4626
PERIOD 1, from 2024-01-01 to 2025-04-07
ALERT! re_auth_thresh =  2
Now we have 330234 first time note authors (all time)
Now we have 172844 first time note authors 2024-01-01 to 2025-04-07
Now we have 134572 notes written by our 3 choosen algorithms
Now we have 95257 notes classified as MISLEADING
Now we have 77741 notes which have enough ratings
--------         No filter for alignment
FINAL total 77741 Notes
of which author writes again =  57520

NB current (final), not Max, score
running variable (x) length 77741
binary outcome (y) length 77741
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:                 75944
Polynomial Order Est. (p):                  1
Polynomial Order Bias (q):                  2
Kernel:                            Triangular
Bandwidth Selection:                    mserd
Var-Cov Estimator:                         NN

                                Left      Right
------------------------------------------------
Number of Observations         66486       9458
Number of Unique Obs.          66393       9456
Number of Effective Obs.        3370       3667
Bandwidth Estimation           0.059      0.059
Bandwidth Bias                 0.103      0.103
rho (h/b)                      0.568      0.568

Method             Coef.     S.E.   z-stat    P>|z|       95% CI
-------------------------------------------------------------------------
```

```
Conventional      -0.006    0.022   -0.262   7.935e-01    [-0.049, 0.038]
Robust                 -        -   -0.577   5.641e-01    [-0.063, 0.035]

MSE-optimal h = 0.0586

Bias-corrected point estimate: -0.0144  95% CI [-0.0634, 0.0345]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):
  h = 0.0293 (0.5x)  tau = -0.0233  95% CI [-0.0979, +0.0389]  N_L/N_R = 1542/1770
  h = 0.0586 (1.0x)  tau = -0.0058  95% CI [-0.0634, +0.0345]  N_L/N_R = 3370/3667
  h = 0.1171 (2.0x)  tau = +0.0196  95% CI [-0.0214, +0.0503]  N_L/N_R = 10056/7030
```